

Review

78%

7

Out of 9 points

0:58

Avg Time Per Question

Your Answers:

Vignette

You are a junior analyst at an asset management firm. Your supervisor asks you to analyze the return drivers for one of the firm's portfolios. She asks you to construct three regression models of the portfolio's monthly excess returns (RET), starting with the following factors: the market excess return (MRKT), a value factor (HML), and the monthly percentage change in a volatility index (VIX). Next you add a size factor (SMB), and finally you add a momentum factor (MOM). Your three models are as follows:

$$\text{Model 1: } \text{RET}_i = b_0 + b_{\text{MRKT}}\text{MRKT}_i + b_{\text{HML}}\text{HML}_i + b_{\text{VIX}}\text{VIX}_i + \varepsilon_i.$$

$$\text{Model 2: } \text{RET}_i = b_0 + b_{\text{MRKT}}\text{MRKT}_i + b_{\text{HML}}\text{HML}_i + b_{\text{VIX}}\text{VIX}_i + b_{\text{SMB}}\text{SMB}_i + \varepsilon_i.$$

Model 3:

$$\text{RET}_i = b_0 + b_{\text{MRKT}}\text{MRKT}_i + b_{\text{HML}}\text{HML}_i + b_{\text{VIX}}\text{VIX}_i + b_{\text{SMB}}\text{SMB}_i + b_{\text{MOM}}\text{MOM}_i + \varepsilon_i.$$

Your supervisor is concerned about conditional heteroskedasticity in Model 3 and asks you to perform the Breusch–Pagan (BP) test. At a 5% confidence level, the BP critical value is 11.07. You run the regression for the BP test; the results are shown in Exhibit 1.

Exhibit 1

Testing for Conditional Heteroskedasticity

Regression Statistics

Multiple *R* 0.25517

R-Squared 0.06511

Regression Statistics

Adjusted <i>R</i> -Squared	0.01317
Standard Error	18.22568
Observations	96

Now the chief investment officer (CIO) joins the meeting and asks you to analyze two regression models (A and B) for the portfolio he manages. He gives you the test results for each of the models, shown in Exhibit 2.

Exhibit 2

Breusch–Godfrey and Durbin-Watson Test Results

	Test Type	Test Statistic	Critical Value	Independent Variable Is Lagged Value of Dependent Variable
Model A	Breusch–Godfrey	12.124	3.927	Yes
Model B	Durbin–Watson	3.088	2.387	No

The CIO also asks you to test a factor model for multicollinearity among its four explanatory variables. You calculate the variance inflation factor (VIF) for each of the four factors; the results are shown in Exhibit 3.

Exhibit 3

Multicollinearity Test Results

Variable	R^2	VIF
X_1	0.748	3.968
X_2	0.451	1.820

Variable	R^2	VIF
X_3	0.942	17.257
X_4	0.926	13.434

1 Multiple Choice 1 / 1 point

Calculate the BP test statistic using the data in Exhibit 1 and determine whether there is evidence of heteroskedasticity.

☐ A. 1.264, so there is no evidence of heteroskedasticity

☒ B. 6.251, so there is no evidence of heteroskedasticity

☐ C. 81.792, so there is evidence of heteroskedasticity

Feedback

B is correct. The BP test statistic is calculated as nR^2 , where n is the number of observations and R^2 is from the regression for the BP test. So, the BP test statistic = $96 \times 0.06511 = 6.251$. This is less than the critical value of 11.07, so we cannot reject the null hypothesis of no heteroskedasticity. Thus, there is no evidence of heteroskedasticity.

2 Multiple Choice 1 / 1 point

Identify the type of error and its impacts on regression Model A indicated by the data in Exhibit 2.

☒ A. Serial correlation, invalid coefficient estimates, and deflated standard errors.

☐ B. Heteroskedasticity, valid coefficient estimates, and deflated standard errors.

☐ C. Serial correlation, valid coefficient estimates, and inflated standard errors.

Feedback

A is correct. The Breusch–Godfrey (BG) test is for serial correlation, and for Model A, the BG test statistic exceeds the critical value. In the presence of serial correlation, if the independent variable is a lagged value of the dependent variable,

then regression coefficient estimates are invalid and coefficients' standard errors are deflated, so t -statistics are inflated.

3 Multiple Choice 1 / 1 point

Determine using Exhibit 3 which one of the following statements is *most likely* to be correct. Multicollinearity issues exist for variables:

- ☐ A. X1 and X2.
- ☐ B. X2 and X3.
- ☒ C. X3 and X4.

Feedback

C is correct. A VIF above 10 indicates serious multicollinearity issues requiring correction, while a VIF above 5 warrants further investigation of the given variable. Since X3 and X4 each have VIFs above 10, serious multicollinearity exists for these two variables. VIFs for X1 and X2 are both well below 5, so multicollinearity does not appear to be an issue with these variables.

4 Multiple Choice 1 / 1 point

Identify the correct answer related to the following statement.

Possible solutions for addressing the multicollinearity issues identified in Exhibit 3 include:

1. excluding one or more of the regression variables.
2. using a different proxy for one of the variables.
3. increasing the sample size.

- ☐ A. Only Solution 1 is correct.
- ☐ B. Only Solution 2 is correct.
- ☒ C. Solutions 1, 2, and 3 are each correct.

Feedback

C is correct. Possible solutions for addressing multicollinearity issues include all of the solutions mentioned: excluding one or more of the regression variables, using a different proxy for one of the variables, and increasing the sample size.

Vignette

Anna Gabor is an analyst with an investment management firm. She is investigating the impact of leverage on annual ROE in the firm's Eurozone equities fund ("the fund").

She collects data on the 72 stocks in the fund, including company size as an independent variable.

Exhibit 1 shows the results from estimating the model $R_i = b_0 + b_1(\text{Size}_i) + b_2(D/E_i) + \varepsilon_i$

Where:

- R_i is the annual ROE for company i in the last calendar year in numbers (0.05 is 5%);
- Size_i is the natural log of the market capitalization of company i (millions of Euro) on 31 December;
- D/E_i is the debt/equity ratio of company i on 31 December;
- b_0 is the intercept of the equation;
- b_1, b_2 are the slope coefficients for the independent variables Size_i and D/E_i respectively;
- ε_i is the error term.

Exhibit 1

	Coefficient	Standard Error
Intercept	0.2248	---
Size_i	-0.012	---
D/E_i	-0.0172	0.00986
ANOVA	Degrees of freedom	Sum of squares
Regression	2	0.0960
Residual	69	0.5704

Exhibit 2 shows selected values of Student's t distribution for 69 degrees of freedom.

Exhibit 2
Selected Values of Student's t distribution

Area in Right Tail	t Value (69 degrees of freedom)
0.05	1.6670
0.025	1.9950

Gabor expects to find that both the debt/equity ratio and size are negatively related to ROE. She considers a relationship meaningful when it is statistically significant at the 0.05

level. As part of her analysis Gabor compares the predictions of her model with the actual ROE for two stocks in the portfolio which have the same debt ratio: A and B, which have market capitalizations EUR 100 million and EUR 200 million respectively.

Gabor shares her results with a colleague, Eva Szabo, who warns that the variance of the error terms might not be constant, but positively correlated with D/E and Size. Szabo states that such a flaw would lead to the F- and t-statistics being biased and failing to reject the null hypothesis when it is false.

As an additional goal Gabor investigates the impact of qualitative governance factors. She is considering adding two dummy variables, to the model: D_1 and D_2 . D_1 is 1 for firms with two or more female board members, and otherwise 0. The value of D_2 is 1 if the audit committee of the board includes only independent directors, and otherwise 0.

Gabor's hypothesis is that gender diversity is positively related to ROE and that an independent audit committee slows the rate at which increasing leverage affects ROE.

Finally, Gabor wishes to test the hypothesis that whether or not a stock in the fund is listed on a US (as well as European) exchange is determined by the same independent variables as above (including the governance factors). She employs a logistic regression with the dependent variable of A, which takes on a value of 1 for a company listed on a US exchange and otherwise 0. For Tekdat, a company in the fund which does not have gender diversity on its board, the probability, based on the model, of being listed on a US exchange is 0.65. The regression coefficient of D_1 in the logistic model is 0.80 and is statistically significant.

5 Multiple Choice 0 / 1 point

Is Szabo's statement about the variances of the error terms correct?

☒ Yes

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Incorrect Answer Feedback: Incorrect, because the problem described, conditional heteroskedasticity, leads to rejection of the null when the null is true, not failure to reject the null when it is false, does not lead to inconsistency of parameter estimators.

☐ No, it is incorrect regarding the F and t statistics being biased

Incorrect Answer Feedback: Incorrect, because bias is indeed a consequence of conditional heteroskedasticity.

When errors are heteroskedastic, the F-test for the overall significance of the regression is unreliable. This unreliability occurs because the mean squared error is a biased estimator of the true population variance given heteroskedasticity. Furthermore, t-tests for the significance of individual regression coefficients are unreliable because heteroskedasticity introduces bias into estimators of the standard error of regression coefficients. When errors are conditional heteroskedastic, the F-test for the overall regression significance is unreliable because the MSE becomes a biased estimator of the true population variance. Moreover, t-tests of individual regression coefficients are unreliable because heteroskedasticity introduces bias into estimators of the standard error of regression coefficients. Thus, in regressions with financial data, the most likely impacts of conditional heteroskedasticity are that standard errors will be underestimated, so t-statistics will be inflated.

☐ No, it is incorrect regarding the null hypothesis

Correct Answer Feedback: Correct, because the problem described, conditional heteroskedasticity, will lead to the commission of Type I errors where the null hypothesis is rejected when it is true. (Not failing to reject it when it is false as Szabo states) does not lead to inconsistency of parameter estimators.

The candidate first needs to identify that the violation of regression assumptions that Szabo has raised is conditional heteroskedasticity. Heteroskedasticity is a violation of the assumption that the variance of error in a regression is constant across observations. Conditional heteroskedasticity occurs when heteroskedasticity in the error variance is correlated with the values of the independent variables in the regression.

The candidate then needs to understand that bias and not inconsistency is a consequence.

Although heteroskedasticity does not affect the consistency of the regression parameter estimators, it can lead to mistakes in inference. Informally, an estimator of a regression parameter is consistent if the probability that estimates of a regression parameter differ from the true value of the parameter decreases as the number of observations used in the regression increases.

6 Multiple Choice 0 / 1 point

Which of the following is a principle for proper regression model specification?

- ☒ The economic reasoning behind the choice of variables should be provided

Correct Answer Feedback: Correct because, consistent with the principle where the model should be grounded in economic reasoning, the model specification should provide the economic reasoning behind the choice of variables.

- ☐ The model should include only those dependent variables which explain the independent variable

Incorrect Answer Feedback: Incorrect because it is the independent variables which explain the dependent variable, and not the other way around. The model should be parsimonious, i.e., each variable included in the regression should play an essential role.

- ☐ If a nonlinear relationship between regressors is expected, the nonlinear terms should be excluded since the model is based on linear regression

Incorrect Answer Feedback: Incorrect because the model functional form should be appropriate, i.e., if a nonlinear relationship between regressors is expected, the model should incorporate the appropriate nonlinear terms, not exclude them.

7 Multiple Choice 1 / 1 point

The null hypothesis in the Breusch–Pagan (BP) test is that there is:

- ☒ no conditional heteroskedasticity.

Correct Answer Feedback: Correct because the Breusch–Pagan (BP) test is widely used in financial analysis to diagnose potential conditional heteroskedasticity and the null hypothesis is that there is no conditional heteroskedasticity.

- ☐ no unconditional heteroskedasticity.

Incorrect Answer Feedback: Incorrect because the Breusch–Pagan (BP) test is widely used in financial analysis to diagnose potential conditional (not unconditional) heteroskedasticity and the null hypothesis is that there is no conditional (not unconditional) heteroskedasticity.

- ☐ neither conditional heteroskedasticity nor unconditional heteroskedasticity.

Incorrect Answer Feedback: Incorrect because the Breusch–Pagan (BP) test is widely used in financial analysis to diagnose potential conditional (not conditional and unconditional) heteroskedasticity and the null hypothesis is that there is no conditional (not conditional and unconditional) heteroskedasticity.

8 Multiple Choice 1 / 1 point

Positive serial correlation in the residuals of a regression model typically results in:

- ☒ more Type I errors.

Correct Answer Feedback: Correct because positive serial correlation does not affect the consistency of regression coefficients, but it does affect statistical tests. First, the F -statistic may be inflated because the MSE (mean squared error) will tend to underestimate the population error variance. Second, positive serial correlation typically causes standard errors to be underestimated, so t -statistics are inflated, which (as with heteroskedasticity) leads to more Type I errors.

- ☐ a deflated F -statistic.

Incorrect Answer Feedback: Incorrect because positive serial correlation affects statistical tests. Specifically, the F -statistic may be inflated (not deflated) because the MSE (mean squared error) will tend to underestimate the population error variance.

- ☐ inconsistent regression coefficients.

Incorrect Answer Feedback: Incorrect because positive serial correlation does not affect the consistency of regression coefficients, but it does affect statistical tests.

9 Multiple Choice 1 / 1 point

Which of the following methods is *most* appropriate to detect multicollinearity in a regression model?

- ☒ Variance inflation factors

Correct Answer Feedback: Correct because we can use the variance inflation factor (VIF) to quantify multicollinearity issues. To understand the size of the multicollinearity problem, the analyst may compute VIFs for each independent variable in a regression. If variables in the regression have large VIFs, they indicate that the error variances of their estimated coefficients are highly inflated and these variables are highly correlated.

- ☐ The Breusch–Pagan test

Incorrect Answer Feedback: Incorrect because a Breusch–Pagan (BP) test is used to detect heteroskedastic error terms, not multicollinearity. The Breusch–Pagan (BP) test is widely used in financial analysis to diagnose potential conditional heteroskedasticity. The null hypothesis for the BP test is that there is no conditional heteroskedasticity.

- ☐ The Breusch–Godfrey test

Incorrect Answer Feedback: Incorrect because a Breusch–Godfrey (BG) test is used to detect serial correlation, not multicollinearity. The BG test is more robust than the Durbin–Watson test because it can detect autocorrelation up to a pre-designated order p , instead of only testing for first-order serial correlation.